

EE227 – Digital Logic Design

Lecture 4 – 5

Binary Storage and Registers

- The binary information in a digital computer must have a physical existence in some medium for storing individual bits
- A **binary cell** is a device that possesses two stable states and is capable of storing one bit (0 or 1) of information
- The input to the cell receives excitation signals that set it to one of the two states
- The output of the cell is a physical quantity that distinguishes between the two states
 - The information stored in a cell is 1 when the cell is in one stable state
 - And 0 when the cell is in the other stable state

Registers

- A register is a group of binary cells
- A register with n cells can store any discrete quantity of information that contains n bits
- The content of a register is a function of the interpretation given to the information stored in it

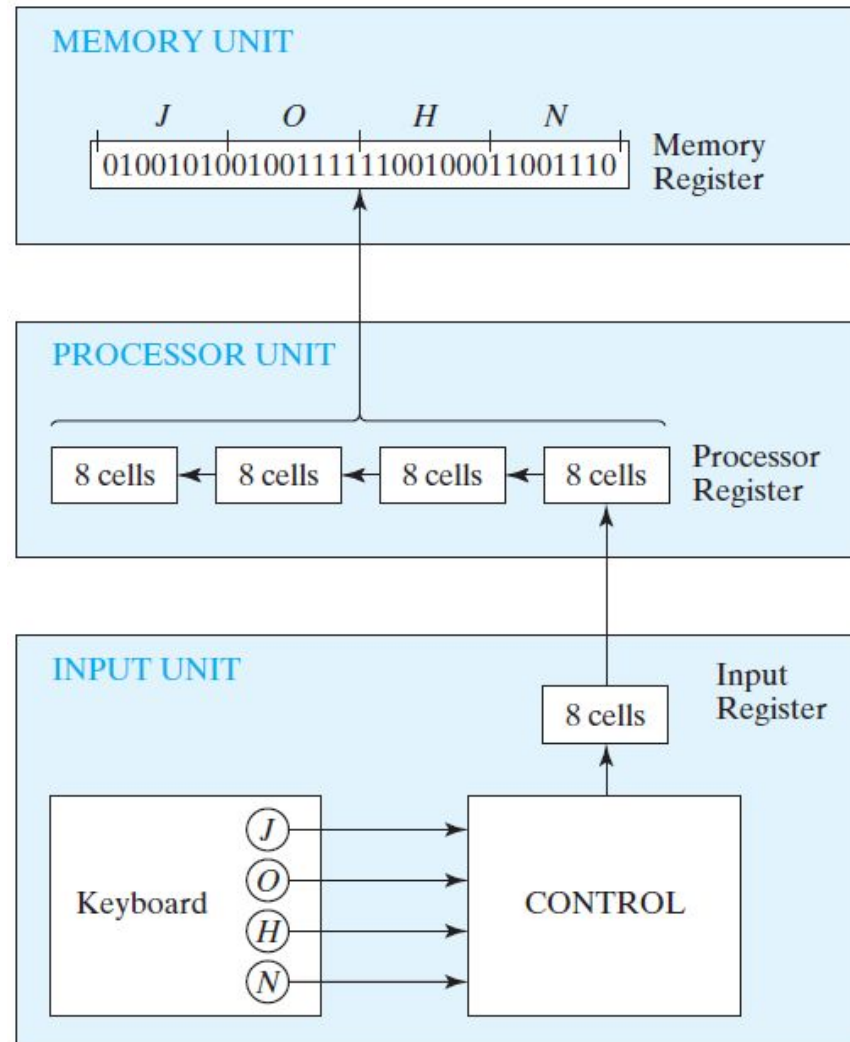
1100001111001001

- The contents of above register are:
 - 50,121 if simple binary number is considered
 - C and I if ASCII with even parity is considered

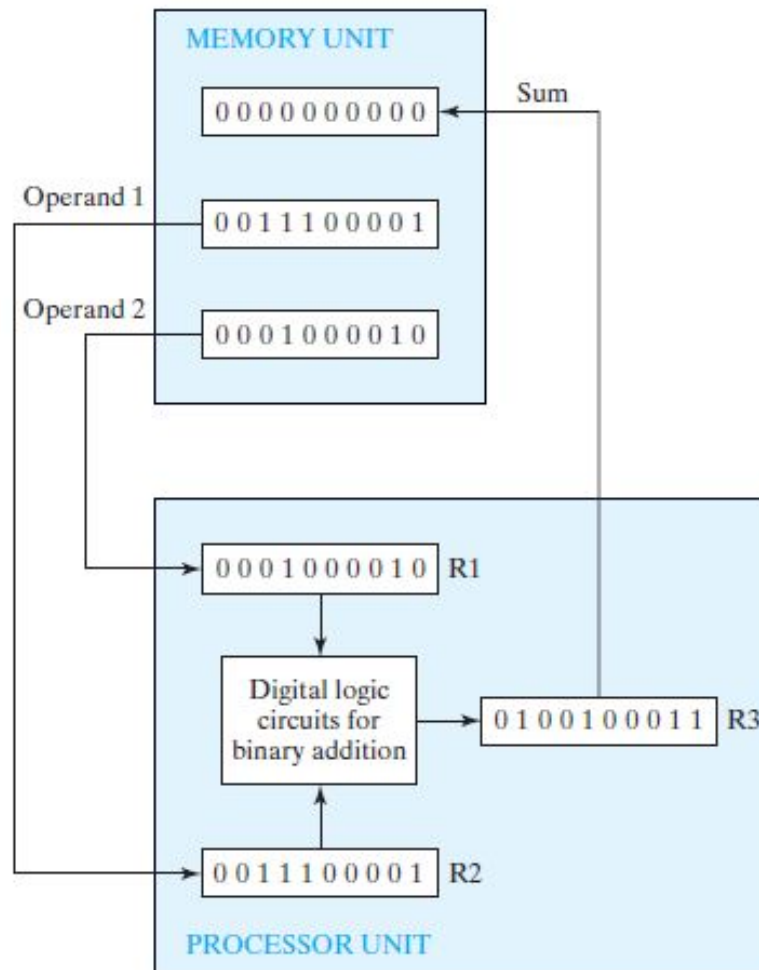
Register Transfer

- In digital systems, a register transfer operation is a basic operation that consists of a transfer of binary information from one set of registers into another set of registers
- The transfer may be direct, from one register to another, or may pass through data-processing circuits to perform an operation
- To process discrete quantities of information in binary form, a computer must be provided with devices that hold the data to be processed
- The device most commonly used for holding data is a register

Transfer of Information from Keyboard to Memory Unit



Binary Information Processing in Adding Two Numbers



Binary Logic

- Binary logic deals with variables that take on two discrete values and with operations that assume logical meaning
- The two values the variables assume may be called by different names (true and false, yes and no, etc.), but for our purpose, it is convenient to think in terms of bits and assign the values 1 and 0

Definition of Binary Logic (1/2)

- Binary logic consists of binary variables and a set of logical operations
- The variable have two and only two distinct possible values: 1 and 0
- There are three basic logical operations: AND, OR, and NOT. Each operation produces a binary result, denoted by z
- AND
 - This operation is represented by a dot or by the absence of an operator
 - For example $x.y = z$ or $xy = z$ both will be read as x AND y is equal to z
 - The logical operation AND is interpreted to mean that $z = 1$ if and only if $x = 1$ and $y = 1$; otherwise $z = 0$

Definition of Binary Logic (2/2)

- OR
 - This operation is represented by a plus sign. For example, $x + y = z$ is read “x OR y is equal to z”
 - Meaning that $z = 1$ if $x = 1$ or if $y = 1$ or if both x and y are 1
 - If both $x = 0$ and $y = 0$, then $z = 0$
- NOT
 - This operation is represented by a prime (sometimes by an overbar). For example, $x' = z$ (or $\bar{x} = z$) is read “not x is equal to z,” meaning that z is what x is not
 - In other words, if $x = 1$, then $z = 0$, but if $x = 0$, $z = 1$

Truth Table

- A truth table is a table of all possible combinations of the variables, showing the relation between the values that the variables may take and the result of the operation
- Truth tables of AND, OR and NOT are shown below

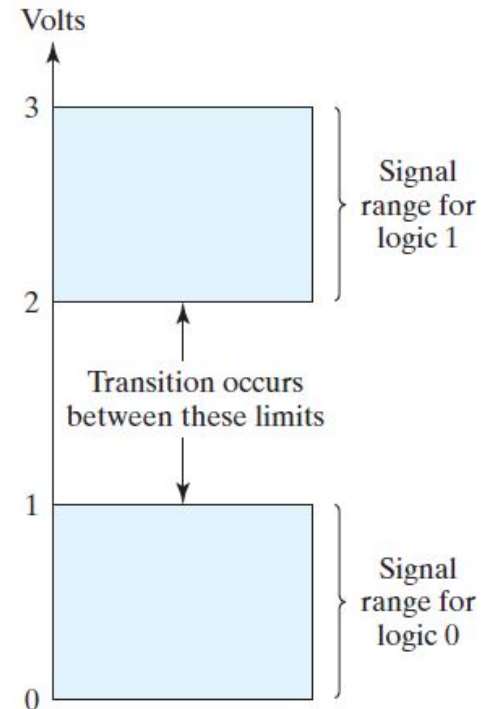
AND			OR			NOT	
x	y	$x \cdot y$	x	y	$x + y$	x	x'
0	0	0	0	0	0	0	1
0	1	0	0	1	1	1	0
1	0	0	1	0	1		
1	1	1	1	1	1		

Binary Arithmetic and Binary Logic

- Binary logic resembles binary arithmetic, and the operations AND and OR have similarities to multiplication and addition, respectively
- In fact, the symbols used for AND and OR are the same as those used for multiplication and addition
- However, **binary logic should not be confused with binary arithmetic**
- One should realize that an arithmetic variable designates a number that may consist of many digits. A logic variable is always either 1 or 0
- For example, in binary arithmetic, we have $1 + 1 = 10$ (read “one plus one is equal to 2”), whereas in binary logic, we have $1 + 1 = 1$ (read “one OR one is equal to one”)

Logic Gates

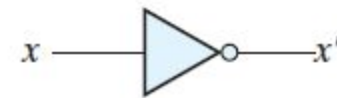
- Logic gates are electronic circuits that operate on one or more input signals to produce an output signal
- The voltages and currents are continuous
- The inputs of gates must be digital
- The conversion from analog to digital is done through defined voltage/current levels
- For example the values greater than a specific threshold will be treated as 1 and values less than that specific threshold are treated as 0
- The graphical symbols of three gates are also shown below



(a) Two-input AND gate



(b) Two-input OR gate



(c) NOT gate or inverter

Timing Diagrams

- The timing diagrams illustrate the idealized response of each gate
- The horizontal axis of the timing diagram represents the time, and the vertical axis shows the signal as it changes between the two possible voltage levels

